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Herataunga Plains Groundwater Study – Volume 1 (1997)

赫利汤加平原地下水研究：卷 1（1997 年）

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Herataunga Plain Groundwater Study: Volume 1- Findings

赫利汤加平原地下水研究：卷 1 - 调查结果

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2.3.8 冰河期后（距今 14000 年）

[...]

赫利汤加平原曾一度被来自陶坡火山区的火山灰覆盖，是由火山尘埃沉降以及由纳如若河和杜塔科尔河冲刷下来的浮石沉积带来的。这些火山沉积物由在平原上钻的水井贯穿。

[...]

Herataunga Plain Groundwater Study- Executive Summary

赫利汤加（Herataunga）平原地下水研究——执行摘要

Introduction – Page 1

简介 – 第 1 页

赫利汤加平原由位于新西兰北岛东海岸的 300 平方公里面积构成，包含黑斯廷斯（Hastings），哈夫洛克北（Havelock North）及内皮尔（Napier）的城区。平原是在过去 25 万年里，由杜塔科尔河（Tutaekuri），纳如若河（Ngaruroro）和图基图基河（Tukituki）的河流沉积物，以及沿海环礁湖、河口和海湾沉淀物形成的。赫利汤加平原的地下层依次是冲积砂砾、沙子和泥沙，中间夹有海洋砂砾、沙子和泥沙渠道。总体为分层结构，由可渗透的粗砂砾层和不可渗透的细沙地层交替构成。可渗透的砂砾层构成了地下蓄水层，在计划中反应出它们的形成是作为蜿蜒的河渠。

[...]

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沉积的地质历史以及地下蓄水系统的深度

三个深度探索钻孔为地质历史及赫利汤加平原下蓄水层的深度提供了信息。

钻孔、岩层及地下水抽样及分析证实了深达 250 米的、有 25 万年历史的冰河期/间冰期岩层结构，中间夹有连通的砂砾蓄层。沉积的出现是由于每 1000 年下降 1 米的下陷。

[...]



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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

The Heretaunga Plains and southern Hawke's Bay (Fig. a) infill a basin (Heretaunga depression-Kingma 1970) about 150 km west of the converging Australian and Pacific plates at the Hikurangi subduction zone (Fig. 1.1). Contrasting tectonic and sedimentation

processes during the Quaternary (last 2 million years) are reflected in folding and thrust faulting, changes in sediment sources as rivers adjusted to tectonic processes, and sedimentation response to global climate cycles and related sea-level changes.

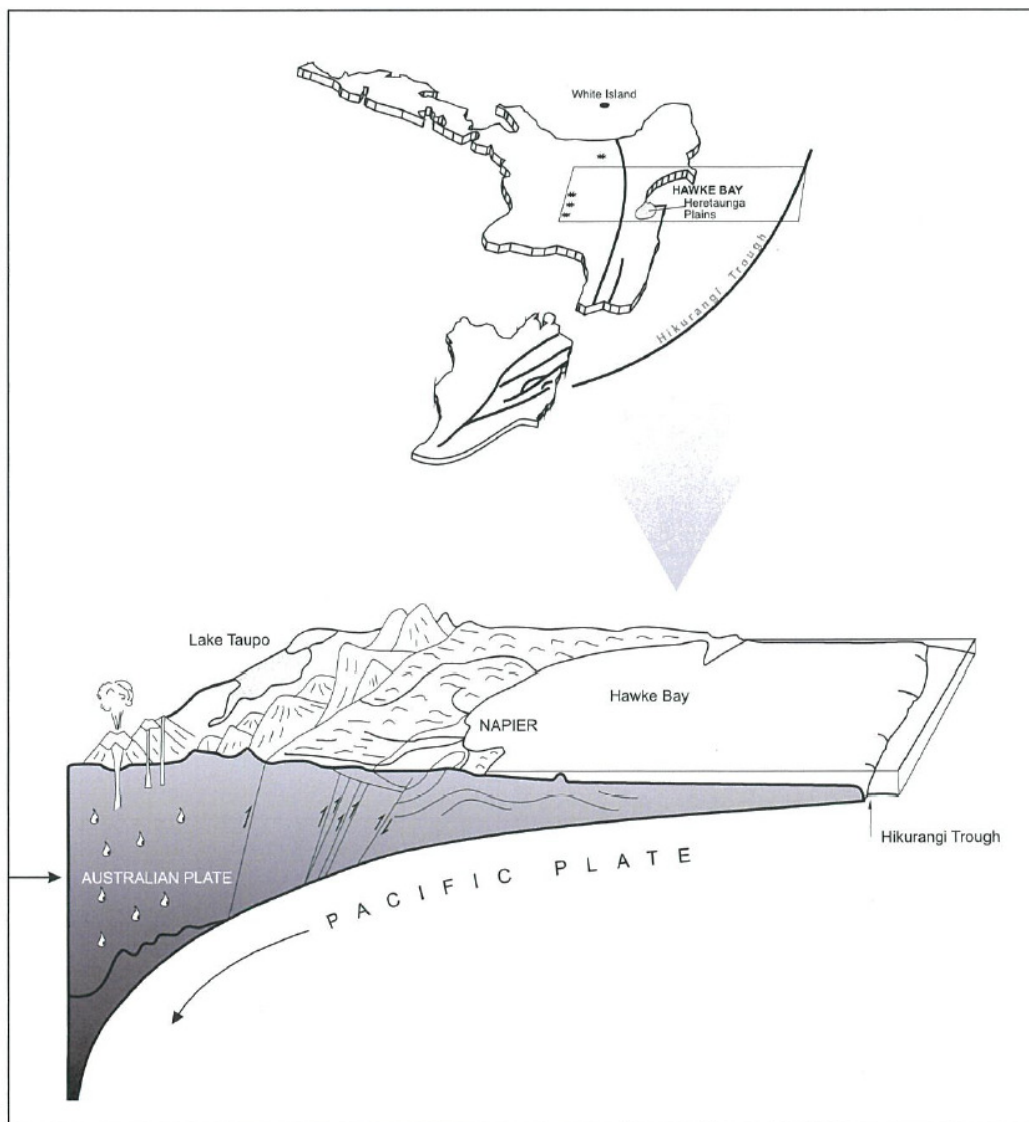


Figure 1.1: Plate tectonic interpretations

hills south of Bridge Pa and at Pakipaki. These gravel fan deposits were constantly reworked and redeposited so that channels of permeable gravel with minimal fines (silt) content criss-crossed the Plains. Poorly sorted gravel, sand and silt formed “islands” where the original fluvial deposits were not reworked and fines, principally silt, accumulated as overbank flood deposits adjacent to the channels. The permeable gravel channels predominated in the central area of the plain, while less permeable fluvial deposits were more likely to be on the margins of the plain. These fluvial processes were typical of the build up of the Heretaunga Plains during the Otiran Glaciation.

About 14 000 years BP when climate warming began, rivers carried less sediment as catchment erosion declined with increased forest cover, and rising sea level shortened the Plains reaches of the river courses. On the Heretaunga Plains the Ngaruroro and Tutaekuri rivers adjusted by downcutting into and reworking the gravel fan and channel deposits. The net result of glacial and early postglacial river deposition and reworking was that the Heretaunga Plain was built up and out with an underlying “spaghetti” of criss-crossing interconnected gravel channels.

2.3.8 Aranuiian Postglacial (c. 14 000 years BP - present day)

About 9000 years BP the rising transgressing sea reached the present day coast - sea level was about 25 m

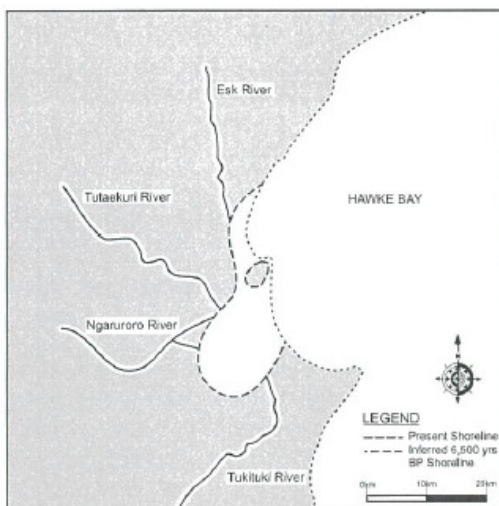


Figure 2.14: Hawke Bay coastline about 6500 years ago.

below present sea level (Fig. 2.20). Sea level continued to rise until about 6500 years BP when the present level was attained. At this time the Hawke Bay coast was inland as far as Pukahu, Longlands, Frimley, Twyford, Taradale and north as marked by the sea cliffs on the western side of Ahuriri Lagoon (Fig. 2.14).

The sandstone and limestone promontory connecting Scinde Island to the mainland had been eroded and Scinde Island along with other smaller islands were present in a bay offshore from Taradale and Bay View. The marine transgression deposited a wedge of marine beach, estuarine and lagoonal sediments and marginal marine swamp deposits over the permeable gravel channel deposits of the last glacial period. At Awatoto this wedge forms a confining strata for the underlying artesian gravel aquifers and is about 30 m thick.

From 6500 years BP to the present, the coast has prograded (built out). About 4500 years BP gravel and sand tombolos had formed to reconnect Scinde Island to the mainland and enclose a large lagoon into which the Tutaekuri and Esk rivers flowed. The Ngaruroro and Tukituki river mouths were at the Hawke Bay coast not far from their present outlets.

For the last 4500 years the lagoon was slowly filled with sediments brought in by the Tutaekuri and Esk rivers and also periods of sediment input by the Ngaruroro River during floods or changes of course. The process was hastened this century by the 1931 earthquake and reclamation by man.

To the south of the Ahuriri Lagoon the Ngaruroro River continued depositing gravel, sand and silt on the Plains in channels changing position in response to tectonic influences and flood events. The Heretaunga Plains has been periodically mantled by tephras originating from the Taupo Volcanic Zone. This has occurred both as airfall deposits and as pumice deposits washed down the Ngaruroro and Tutaekuri rivers. These volcanic deposits are intersected by wells drilled on the Heretaunga Plains. Taupo pumice derived from the Taupo eruption about 140 AD, centred on what is now Lake Taupo is overlain by up to 10 m of fluvial deposits. A former Ngaruroro River course in the Bridge Pa - Pakipaki area was clogged with Taupo pumice shortly after the eruption and adjacent areas of the Heretaunga Plains were capped by pumice overbank flood deposits.

CHAPTER I

INTRODUCTION

The Heretaunga Plains comprise an area of 300 km² on the east coast of the North Island of New Zealand (Fig. 1), incorporating the urban areas of Hastings, Havelock North and Napier. The Plains were formed during the last 250 000 years by river sediments deposited by the Tutaekuri, Ngaruroro and Tukituki rivers and coastal lagoon, estuarine and embayment deposits. The Heretaunga Plains are underlain by sequences of fluvial gravel, sand and silt, interbedded with marine gravel, sand and silt channels. There is a general layered structure with coarse permeable gravel beds alternating with fine impermeable beds. The permeable gravel beds form aquifers which in plan reflect their formation as meandering river channels.

The groundwater underlying the Heretaunga Plains is a major natural resource which provides about 85% of the Heretaunga Plains and adjacent areas water requirements. Approximately 63 million cubic metres of groundwater was withdrawn during the year July 1994-June 1995 period for public water supply, irrigation and industrial uses. Figure 2 indicates the water used in three principal sectors.

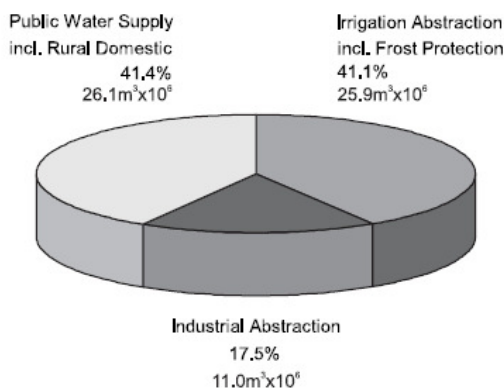


Figure 2: Heretaunga Plains groundwater use during July 1994-June 95

Demand for irrigation from groundwater has expanded considerably in the last twenty years. The combination of temperate climate, fertile soil and groundwater for irrigation, enables the production of about 50% of the total New Zealand harvest of fruit, vegetables and grapes

on the Heretaunga Plains. During dry periods, agricultural and horticultural production is heavily reliant on irrigation from groundwater supplies. Currently there are 2500 resource consents for groundwater abstraction on the Heretaunga Plains, of which 2420 are for irrigation.

Since the discovery of artesian groundwater in the Heretaunga Plains in 1867 numerous groundwater investigations have been undertaken to determine how the aquifers were formed, the sources and flow of water in the aquifers and, in particular, susceptibility to contamination. While these investigations extended groundwater knowledge considerably, it was not until 1991 that a coordinated and systematic regional investigation began. The impetus for this renewed research and investigation was provided by a significant, new legislation. The Water and Soil Conservation Act (1967) was replaced by the Resource Management Act (RMA) in 1991. The new legislation required a far wider perspective of environmental management, with a specific focus on the effects of activities.

STATUTORY CONTEXT

The RMA has as a core requirement, the promotion of sustainable management of physical and natural resources. The RMA places specific emphasis on:

- sustaining the potential of natural and physical resources;
- safeguarding the life-supporting capacity of air, soil, water and ecosystems;
- avoiding, remedying or mitigating any adverse effects of activities on the environment.

The Hawke's Bay Regional Council (HBRC) has identified objectives and priorities for achieving sustainable resource management in the Hawke's Bay Region in the Regional Policy Statement (RPS), which provides direction for a comprehensive Regional Water Resources Plan (RWRP). The Heretaunga Plains groundwater study will provide additional technical input to future reviews of the RWRP.

CHAPTER 3

INVESTIGATIONS AND RESULTS

Review and collate the existing groundwater information and identify the gaps in knowledge.

Significant “gaps in knowledge” identified were:

- the geological history of deposition and the depth of the aquifer sequence beneath the Heretaunga Plains;
- the groundwater flow direction and flow rates;
- the spatial and time aspects of groundwater chemistry and influence of landuse activities and aquifer lithologies;
- total groundwater recharge, abstractions and out-flow;
- the vulnerability to pollution and the continuity of groundwater availability of the aquifers at the fringe of the Heretaunga Plains aquifer system.

GEOLOGICAL HISTORY OF DEPOSITION AND THE DEPTH OF THE AQUIFER SYSTEM.

Three deep exploratory bores were drilled to provide information on the geological history and the depth of the aquifer sequence beneath the Heretaunga Plains.

The drilling, and strata and groundwater sampling and analyses, established that a 250,000 years glacial / interglacial strata sequence was present to a depth of 250 m

with interbedded and interconnected gravel aquifers. Deposition had occurred in a depression subsiding at a rate of 1 m per 1000 years.

Figure 4 illustrates a generalised cross section through the three deep exploratory bores and offshore Hawke Bay. The Flaxmere testbore located in the unconfined aquifer area encountered water bearing (saturated) gravels to the final depth of 137 m. The Tollemache Orchard bore located in about the middle of the Heretaunga depression was explored to a final depth of 256 m. Five flowing artesian and one subartesian aquifers were encountered. The coastal Awatoto testbore penetrated six flowing artesian aquifers. The piezometric pressures in Tollemache Orchard testbore show gradual decrease with depth whereas the piezometric pressures increase with depth at the Awatoto testbore. Because of the shallow depositional gradient of the aquifers, only the upper gravel aquifers are likely to outcrop on the sea floor and discharge fresh water into the ocean in Hawke Bay.

The decreasing piezometric pressures observed at the Tollemache Orchard testbore and other deeper bores in the inland Plains, support the concept of a hydraulically interconnected system with a common groundwater recharge source area. Increasing piezometric pressures with depth in the aquifers near the coast suggest a closed or partly closed deeper aquifer system with flow in deeper aquifers maintained by upward leakage through aquitards and aquicludes.

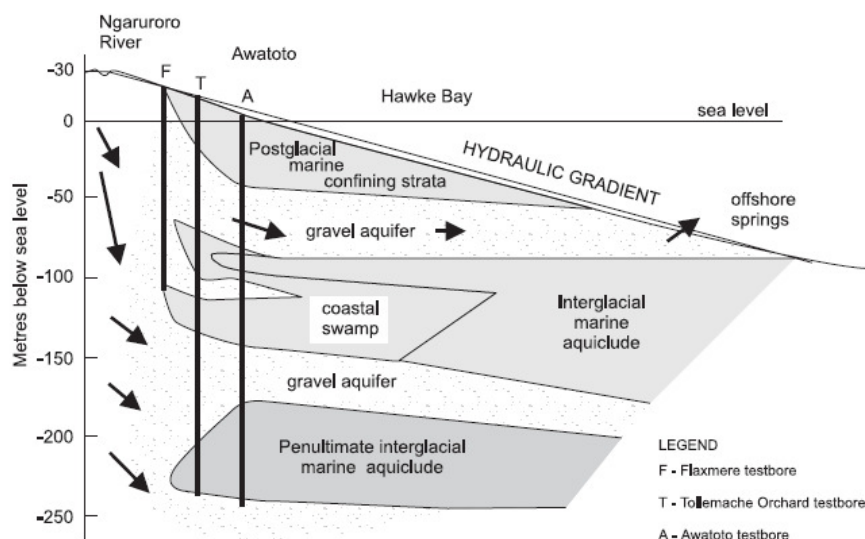


Figure 4: Generalised cross section showing exploratory bores and off-shore Hawke's Bay.

